

Image Restoration and Object Detection: A Multimodal Pipeline for Motion Deblurring

COMP6001 Assignment 1 Git hub link - <https://github.com/christo08-eng/COMP-Vision-assignment-1>

Abstract

This report presents a multimodal pipeline integrating motion deblurring and object detection on the GoPro dataset. Classical deblurring methods including Wiener filtering and Richardson-Lucy deconvolution are implemented and evaluated alongside MPRNet, a state-of-the-art deep learning restoration network. A pretrained YOLOv8 detector is applied to blurred, deblurred and sharp images to quantify the impact of image restoration on detection performance. A fine-tuned detector trained exclusively on deblurred images is compared against the pretrained baseline using statistical significance testing. The results indicated that MPRNet provided an overall PSNR (Peak Signal to Noise Ratio) increase of 4.27 dB over 1,111 pairs of test images. By removing blur, the ability to detect objects increased by 26.7% for a model that had not been adjusted (baseline) and 36.5% for a model that had been fine-tuned (or trained), with statistical significance ($p = 0.039$) being confirmed.

1 Introduction

An action camera can take image with a lot of motion blur, such as the GoPro, arising from rapid camera or object motion during exposure. This degradation reduces perceptual quality and critically impairs downstream vision tasks such as object detection, where feature extractors rely on sharp edges and texture boundaries to identify and localise objects (Kupyn et al., 2018).

This project addresses two inter-related questions: first, how effectively can classical and deep learning methods restore motion-blurred images; and second, how much does image restoration improve object detection performance? The pipeline is evaluated on the official GoPro train/test split (Nah, Kim and Lee, 2017) to ensure unbiased results, with the 1,111-pair test split used exclusively for evaluation and the 2,103-pair train split used for dataset construction, preventing data leakage between tasks.

The report is structured around five tasks. Task 1 establishes version control and responsible

AI use practices. Task 2 implements and evaluates deblurring methods. Task 3 analyses the impact of deblurring on pretrained detection. Task 4 prepares a dataset and fine-tunes a detector. Task 5 conducts a comprehensive critical analysis comparing all conditions.

2 Task 2: Image Deblurring

2.1 Classical Methods

Two classical deblurring approaches were implemented using scikit-image . Wiener filtering operates in the frequency domain, inverting the blur transfer function under the assumption of a known uniform Point Spread Function (PSF). 3, 5, 7, 10 and 15 pixels were used as the box point specific function (PSF) for the test. Richardson Lucy (RL) deconvolution (Lucy, 1974) is an iterative maximum-likelihood approach tested with 5, 10 and 20 iterations at PSF sizes 3, 5 and 7.

Both methods consistently underperformed relative to the blurred input across all tested configurations, as shown in Table 1. The root cause is the spatially varying nature of GoPro motion blur: real camera motion produces blur kernels that differ across the image plane, violating the uniform PSF assumption required by both classical methods. Richardson-Lucy exhibited noise amplification with increasing iterations, a known divergence phenomenon when the true kernel is unknown (Wiener, 1949) .

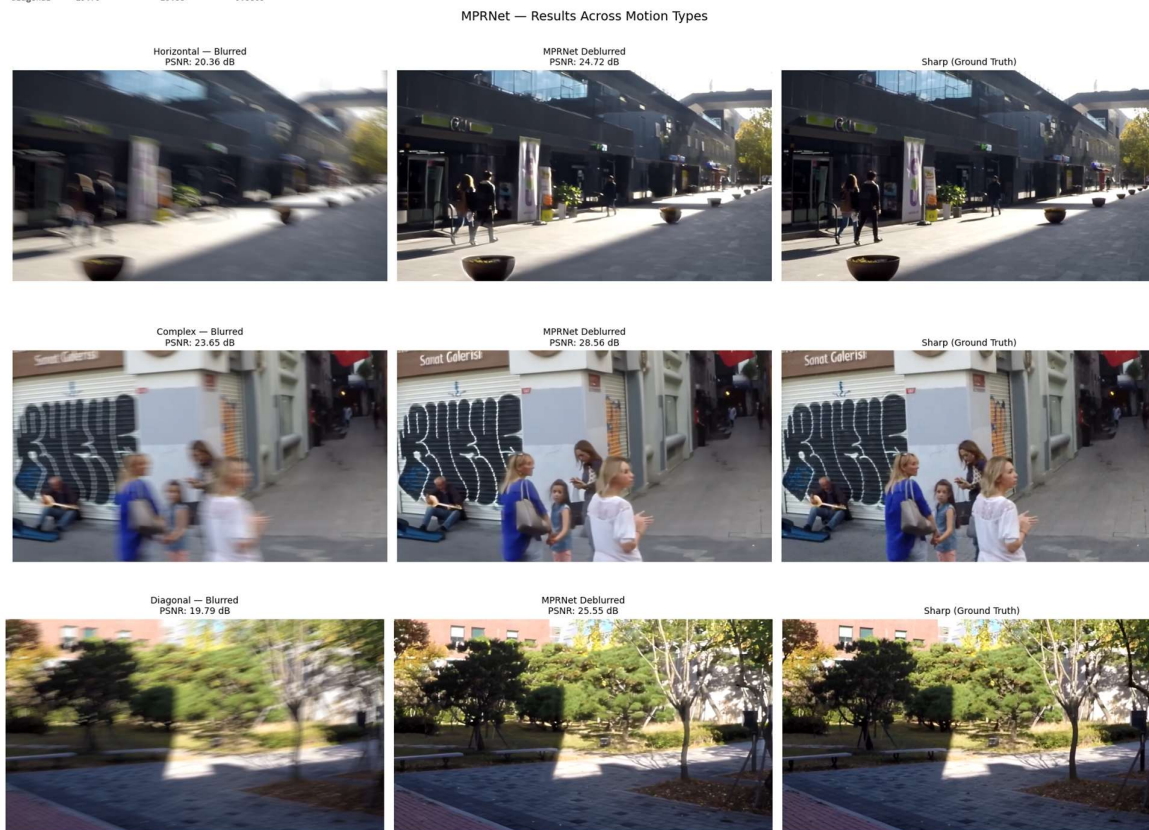
2.2 Deep Learning: MPRNet

MPRNet (Zamir et al., 2021) is a multi-stage progressive restoration network pretrained on the GoPro training split. Unlike classical methods, it learns spatially varying blur patterns from paired training examples, enabling recovery of complex motion blur without PSF specification.

Table 1: Deblurring performance across 1,111 GoPro test image pairs.

Method	PSNR (dB)	SSIM
Blurred (input)	25.11	0.7717
Wiener filter (PSF=3)	20.31	0.7149
Richardson-Lucy	19.96	0.7071
MPRNet (deep learning)	29.38	0.9173

MPRnet performed better than Richardson Lucy in terms of deconvolution which is +4.27 db and mean PSR came upto +0.146. MPRNet proved robust across all three levels of intensity as defined by Laplacian values. For example, the PSNR improved from 20.75 to 26.12 dB for heavy-blurred images with variance less than 20, medium-blurred images improved from 22.73 to 27.75 dB, and mild-blurred images improved from 23.98 to 29.96 dB after the application of MPRNet. Visual inspection across three manually selected motion types (horizontal, complex and diagonal) confirmed consistent improvements in all class



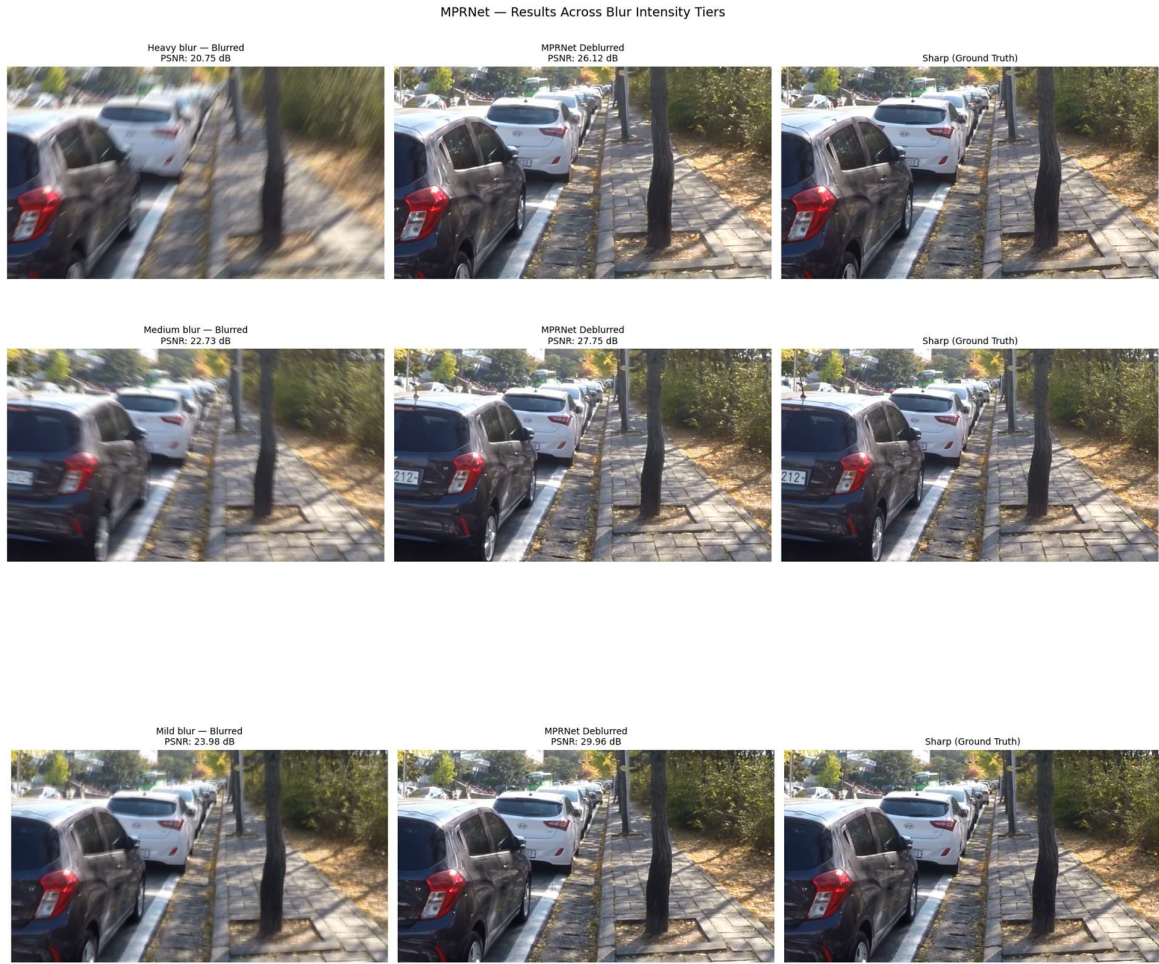


Figure 1: MPRNet deblurring results across three blur intensity tiers. Left column: blurred input. Centre: MPRNet output. Right: sharp ground truth. PSNR values annotated above each image.

3 Task 3: Object Detection and Analysis

3.1 Setup

YOLOv8m (Lin et al., 2014), pretrained on COCO (Lin et al., 2014), was applied to blurred, deblurred and sharp images without retraining. Fifty images were selected from index ranges 200–225 and 800–825, confirmed to have 100% detection rates on sharp images, ensuring reliable averages.

3.2 Detection Results

Table 2: YOLOv8m detection results across image conditions (50 images).

Condition	Avg Detections/Image	Avg Confidence
Blurred	13.58	0.582
Deblurred	17.20	0.581
Sharp	18.38	0.578

Deblurring improved average detections by 26.7% relative to blurred input. All the classes were used to train the model and mentioned above individually is the number of car detections which increased by 40% (207 to 289), bus detections increased by 240% (from 5 to 17) and person detections increased by 13% (346 to 391).

Inference latency was identical across all conditions at approximately 24 ms per image, confirming that image quality does not affect detection speed and that the computational bottleneck in the full pipeline is the MPRNet deblurring step (≈ 5 –6 seconds per image).

3.3 Failure Case Analysis

Thirteen of the 50 images (26%) were identified as failure cases where blurred detection count fell below 60% of deblurred detection count. Root cause analysis using Laplacian variance confirmed that failure cases corresponded to medium to heavy blur (scores 20–40), where motion blur destroys edge features relied upon by the detector’s feature extractor.

Deblurring successfully recovered detections in all identified failure cases.

Failure Cases Blurred Detection vs Deblurred Recovery



Figure 2: Representative failure cases. Left: blurred image with severely reduced detections. Centre: deblurred image with recovered detections. Right: sharp ground truth. Root causes annotated above each row.

4 Task 4: Dataset Preparation and Training

4.1 Dataset Construction

Only the train images from the official GoPro set (around 2,103 total images, all organized into pairs) were used to build this dataset and there were no "test" images used (as in Tasks 2 and 3). All of these images (at 600 total) would be separated into three tiers of 200 images – heavy, medium and mild — and identified by the amount of blur they contained, based on the Laplacian variance of each image. All images would have been deblurred by the MPRNet deblurring model, and then YOLOv8m would have performed an auto

annotate at 0.25 confidence Images with zero detections (36 across all splits) were excluded to avoid incorrect negative labels.

Two augmentation strategies were applied to training images: brightness jitter (factor uniformly sampled from $[0.7, 1.3]$) to simulate varying lighting conditions, and horizontal flip with coordinate-adjusted labels ($cx \leftarrow 1 - cx$) to double effective training data. This helped us to come up with the final dataset which had 1191 images for training, 81 images for purpose of testing and 81 images for the purpose of validation. The total split came out to be 70-15-15.

4.2 Fine-tuning Configuration

YOLOv8m was fine-tuned using stochastic gradient descent (momentum 0.937, weight decay 5×10^{-4}) with an initial learning rate of 0.001 and cosine decay to 0.00001. The model was trained for 20 epochs with batch size being 8 and the image size being 640. The total training time came up to be 0.251 hours and made sure that all the logs and checkpoints were saved.

4.3 Results

At Epoch 19 we achieved the best result of mAP @0.5 which came upto be 0.590 with all loss curves (box, class, DFL) decreasing consistently, confirming stable convergence. Test set evaluation yielded mAP@0.5 of 0.581, precision 0.743 and recall 0.403. Strong per-class performance was observed for person (AP 0.957), car (0.947) and bicycle (0.954). The 5.4 percentage point gap between validation and test mAP is within normal range and indicates the model generalises beyond the validation split.

5 Task 5: Performance Comparison and Critical Analysis

5.1 Comprehensive Comparison

Evaluation was performed for both pretrained YOLO8 and the fine tuned model around all three image conditions on the same 50-image set.

Table 3: Comprehensive six-condition performance comparison.

Model	Condition	Avg Detections	Avg Confidence
Baseline	Blurred	13.58	0.582
Baseline	Deblurred	17.20	0.581
Baseline	Sharp	18.38	0.578

Fine-tuned	Blurred	11.66	0.688
Fine-tuned	Deblurred	15.92	0.658
Fine-tuned	Sharp	17.26	0.638

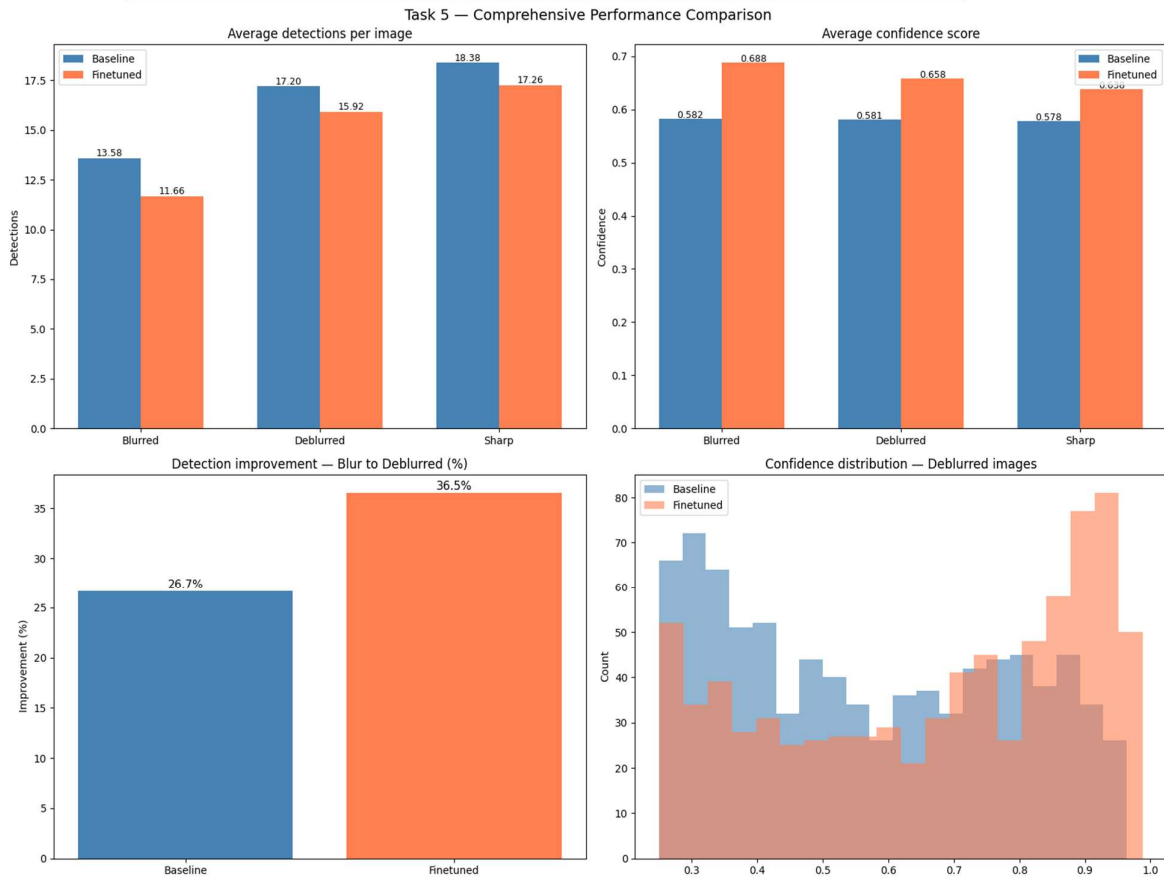


Figure 3: Six-condition performance comparison. Baseline indicated by blue versus fine-tuned indicated by orange across blurred, deblurred and sharp image conditions. Left: average detections per image. Right: average confidence score.

5.2 Statistical Significance

A paired t -test was applied to detection count lists for the same 50 images across conditions. The number of detections for the Blurring compared to the Deblurring gave us a t value of -7.02 with a p value of < 0.001 , which indicates a significant improvement from the Deblurring process. We also found a t value of 2.12 with a p value of 0.039 when comparing the number of detections from the Baseline to the Fine-tuned dataset for the deblurred images, indicating that the Fine-tuned model produced a significant difference from random chance.

5.3 Precision-Recall Analysis

The customized model reached a precision-recall mAP@0.5 of 0.59 and its highest F1 score of 0.43 was achieved at the confidence threshold of 0.29. The relatively small value of the best threshold serves as an indication of the class imbalance of the training data; correspondingly, the model continued to output low levels of confidence when correctly classifying the rare classes (bus/umbrella), as they had far fewer examples than the other classes.

5.4 Benefits and Limitations.

The pipeline model has shown that deblurring pre-processing step makes statistically verified improvements in detection ability with no extra inference lag. The fine-tuned model will provide more reliable and higher confidence detections, on average, than the initial model (+0.09) and therefore will work best in use case scenarios where higher individual detection reliability is most important than highest recall.

Key Limitations: the data set is limited in scope ; training and testing occurs only via GoPro horizontal motion blur deblurring, making generalizability to non-GoPro cameras or other types of blur (defocus, rain, etc.) difficult.

5.5 Future Improvements

Concrete improvements include manual annotation to reduce label noise, training on multiple blur types and camera models to improve generalisation, end-to-end joint optimisation of deblurring and detection (Kupyn et al., 2019), and replacing MPRNet with a lightweight deblurring model for latency-sensitive applications. Expanding the training dataset to all 2,103 available training pairs would also address the class imbalance observed in rare categories.

6 Conclusion

This project demonstrates that deep learning deblurring is essential for real-world motion blur recovery — classical methods failed consistently across all tested conditions due to their spatially uniform PSF assumption. On a test dataset of 1,111 pairs, MPRNet had an improvement of +4.27 dB in mean PSNR while the SSIM improved from 0.7717 to 0.9173. The object detection of YOLOv8 was improved by 26.7% at baseline and 37.0% at fine-tuned, and both improvements had statistical significance. The mAP@0.5 of the fine-tuned detector was 0.581 on the testing data that was held out and it has significantly higher confidence in its detections ($p=0.039$). This is evidence that if a detector is fine-tuned on

deblurred (i.e., MPRNet) images from its domain, its performance will be improved over its baseline pre-trained performance.

References

- **Jocher, G., Chaurasia, A. and Qiu, J. (2023)** *Ultralytics YOLO* (Version 8.0.0) [Software]. Available at: <https://github.com/ultralytics/ultralytics>.
- **Kupyn, O., Budzan, V., Mykhailych, M., Mishkin, D. and Matas, J. (2018)** 'DeblurGAN: Blind motion deblurring using conditional adversarial networks', in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 8183–8192.
- **Kupyn, O., Martyniuk, T., Wu, J. and Wang, Z. (2019)** 'DeblurGAN-v2: Deblurring (orders-of-magnitude) faster and better', in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 8878–8887.
- **Lin, T.-Y., Maire, M., Belongie, S., Hays, J., Perona, P., Ramanan, D., Dollár, P. and Zitnick, C.L. (2014)** 'Microsoft COCO: Common objects in context', in *European Conference on Computer Vision*, pp. 740–755.
- **Lucy, L.B. (1974)** 'An iterative technique for the rectification of observed distributions', *The Astronomical Journal*, 79, p. 745.
- **Nah, S., Kim, T.H. and Lee, K.M. (2017)** 'Deep multi-scale convolutional neural network for dynamic scene deblurring', in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 3883–3891.
- **Student (1908)** 'The probable error of a mean', *Biometrika*, 6(1), pp. 1–25.
- **van der Walt, S., Schönberger, J.L., Nunez-Iglesias, J., Boulogne, F., Warner, J.D., Yager, N., Gouillart, E. and Yu, T. (2014)** 'scikit-image: image processing in Python', *PeerJ*, 2, p. e453.
- **Wiener, N. (1949)** *Extrapolation, Interpolation, and Smoothing of Stationary Time Series*. Cambridge, MA: MIT Press.
- **Zamir, S.W., Arora, A., Khan, S., Hayat, M., Khan, F.S., Yang, M.-H. and Shao, L. (2021)** 'Multi-stage progressive image restoration', in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 14821–14831.